

# **Predicting HIV-1 Nucleoside Reverse Transcriptase Inhibitor (NRTI) Resistance using RT Mutation Patterns and Interpretable Machine Learning**

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## **INTRODUCTION**

Human Immunodeficiency Virus type 1 (HIV-1) remains one of the world's leading infectious diseases, with an estimated 39–40 million people living with HIV globally. Although the widespread scale-up of antiretroviral therapy (ART) has transformed HIV infection into a manageable chronic disease, the emergence and transmission of HIV drug resistance (HIVDR) continues to threaten treatment success. In sub-Saharan Africa (SSA), the HIV burden remains highest (UNAIDS, 2024), with an increasing prevalence of resistance to first-line antiretroviral (ARV) drugs. This poses a significant challenge to achieving the UNAIDS 95-95-95 targets.

Kenya has made substantial progress in expanding ART coverage and viral load monitoring. The widespread availability of ART has transformed HIV infection from a fatal disease into a manageable chronic condition. However, prolonged exposure to antiretroviral drugs has created selective pressure that favours the emergence of HIV drug resistance (HIVDR), threatening individual treatment outcomes and the sustainability of national HIV treatment programmes (WHO, 2024). HIVDR can lead to virological failure, accumulation of additional resistance mutations, increased transmission of resistant virus, and reduced effectiveness of available treatment options (WHO, 2024).

Aims include to develop and interpret machine learning models for predicting NRTI resistance among HIV-1 Group M subtype A viruses using reverse transcriptase mutation profiles obtained from the Stanford HIV Drug Resistance Database (SHIVRDB). We will identify RT mutations associated with NRTI resistance, evaluate the performance of different ML algorithms, determine which models provide the optimal balance between predictive accuracy and interpretability, and assess whether ML-derived predictors correspond with established biological mechanisms of resistance.

Goal: Predict HIV NRTI drug resistance from viral mutation profiles.

Inputs: Mutation features (e.g., M184V, K65R, D67N)

Outputs: Resistance labels (Resistant/susceptible).

## **HIV Treatment in Kenya**

Antiretroviral therapy (ART) is the cornerstone of HIV-1 treatment and has transformed HIV infection from a life-threatening disease into a manageable chronic condition. ART consists of combinations of antiretroviral drugs that target different stages of the HIV-1 replication cycle, with the primary goals of achieving sustained viral suppression, restoring immune function, and reducing HIV-related morbidity and mortality. In Kenya, the national ART programme

initially relied predominantly on regimens combining two nucleoside reverse transcriptase inhibitors (NRTIs) with a non-nucleoside reverse transcriptase inhibitor (NNRTI). For many years, commonly used first-line regimens included zidovudine (AZT) or tenofovir (TDF) combined with lamivudine (3TC) and either efavirenz (EFV) or nevirapine (NVP). With subsequent updates to national treatment guidelines, tenofovir/lamivudine/dolutegravir (TLD) became the preferred first-line regimen for adults, reflecting the broader shift towards dolutegravir-based ART.

HIV-1 infection begins when the virus attaches to the CD4 receptor and appropriate co-receptors on susceptible host cells, followed by membrane fusion and viral entry. Once inside the cell, the viral enzyme reverse transcriptase (RT) converts the viral RNA genome into DNA. The viral DNA is subsequently integrated into the host genome by integrase, after which viral genes are expressed, proteins are produced and assembled, and mature infectious virions are released to infect additional cells. These stages provide multiple targets for antiretroviral therapy, including entry inhibitors, fusion inhibitors, CCR5 antagonists, reverse transcriptase inhibitors, integrase strand transfer inhibitors (INSTIs), protease inhibitors, capsid inhibitors, and maturation inhibitors.

The present study focuses specifically on the reverse transcription stage, where NRTIs exert their antiviral activity. NRTIs, including tenofovir, lamivudine, abacavir (ABC), and zidovudine, are nucleoside or nucleotide analogues that interfere with viral DNA synthesis. Following intracellular activation, these drugs compete with naturally occurring nucleotides and are incorporated into the growing viral DNA strand by HIV-1 reverse transcriptase. Because they lack the structural group required for subsequent nucleotide incorporation, their incorporation results in premature termination of viral DNA synthesis, thereby inhibiting viral replication. The effectiveness of these drugs, however, can be compromised by genetic variation within the HIV-1 reverse transcriptase gene. Selection of drug-resistant variants during ART exposure can result in mutations that reduce susceptibility to one or more NRTIs and may consequently compromise treatment effectiveness.

Several RT mutations have established associations with NRTI resistance. For example, M184V is strongly associated with reduced susceptibility to lamivudine, while K65R is associated with reduced susceptibility to tenofovir and other NRTIs. Thymidine analogue mutations (TAMs), including mutations such as M41L and T215Y, can reduce susceptibility to zidovudine and other NRTIs through altered recognition and excision of incorporated analogue molecules. The occurrence of these mutations, individually or in combination, can therefore generate diverse resistance profiles and influence the effectiveness of commonly used antiretroviral regimens.

Although ART remains highly effective, prolonged treatment exposure, suboptimal adherence, and ongoing viral replication in some individuals can provide conditions for the emergence and selection of drug-resistant HIV-1 variants. Continued surveillance and characterization of HIV-1 drug resistance are therefore important for informing treatment strategies and maintaining the effectiveness of available ART options. Conventional resistance assessment based on genotypic or phenotypic testing can provide valuable information but may not fully capture the complex relationships between combinations of RT

mutations and drug susceptibility. Consequently, machine learning approaches offer an opportunity to identify complex mutation patterns and develop predictive models of NRTI resistance. In this study, HIV-1 reverse transcriptase mutation profiles were therefore investigated to characterize resistance-associated patterns and evaluate machine learning models for predicting phenotypic resistance to lamivudine (3TC), tenofovir (TDF), abacavir (ABC), and zidovudine (AZT).

### **HIV Drug Resistance**

The HIV-1 reverse transcriptase (RT) enzyme is encoded by the pol gene and is essential for conversion of the viral single-stranded RNA genome into double-stranded DNA, which is subsequently integrated into the host genome. Unlike human DNA polymerases, HIV-1 RT lacks proofreading activity. Therefore, errors are introduced during each replication cycle, generating a diverse population of genetically related viruses known as quasispecies. Most mutations are neutral or deleterious; however, some mutations provide a survival advantage when the virus is exposed to antiretroviral drugs. RT gene mutations within its coding region can directly alter drug susceptibility.

**Fold change (FC)** describes how much the susceptibility of a resistant virus differs from a reference susceptible virus.

Calculated as:

*Fold Change = Drug concentration required to inhibit resistant virus/Drug concentration required to inhibit wild-type virus.*

E.g:

*Wild-type virus: IC<sub>50</sub> = 0.1 µM*

*Mutant virus: IC<sub>50</sub> = 10 µM*

*Fold change: 10/0.1=100*

*Therefore the mutant virus requires 100 times more drug concentration for equivalent inhibition.*

### **Rationale**

Despite major advances in HIV treatment and the widespread availability of effective ART, HIV drug resistance (HIVDR) remains an important challenge to sustained treatment success. The emergence and selection of resistant viral variants can compromise the effectiveness of NRTI-containing regimens, resulting in virological failure and limiting future treatment options. This is particularly important in settings such as Kenya, where large-scale ART programmes rely on durable treatment responses to maintain population-level viral suppression.

HIVDR also has direct implications for achieving the UNAIDS 95-95-95 targets, particularly the third 95, which seeks to ensure that people receiving ART achieve viral suppression. Failure to recognize resistance promptly may delay appropriate regimen modification, prolong exposure to ineffective treatment, and increase the potential for onward transmission of resistant virus. There is therefore a need for approaches that can facilitate timely identification and characterization of resistance-associated patterns.

Although genotypic and phenotypic resistance testing remain important tools for HIVDR

surveillance and clinical management, the relationship between combinations of viral mutations and drug susceptibility is complex. Conventional approaches may not readily capture higher-order interactions among mutations that contribute to resistance phenotypes. Furthermore, the increasing availability of HIV sequence and phenotypic susceptibility data provides an opportunity to apply computational approaches to identify these complex patterns.

This is where our study becomes important.

Machine learning (ML) offers a potential approach for predicting drug resistance directly from HIV-1 mutation profiles. However, predictive performance alone is insufficient for applications in biomedical research and clinical decision-making; models should also provide insight into the biological features underlying their predictions. Interpretable ML can therefore provide both predictive information and identification of mutations that contribute most strongly to resistance classification, improving model transparency and facilitating biological interpretation.

The present study addresses this need by investigating whether HIV-1 reverse transcriptase mutation profiles can be used to predict phenotypic resistance to NRTIs. Specifically, the study evaluates resistance to lamivudine (3TC), tenofovir (TDF), abacavir (ABC), and zidovudine (AZT) and examines the mutation patterns associated with the observed resistance phenotypes. By combining mutation profiling, unsupervised exploration, and interpretable supervised machine learning, the study aims to provide a data-driven framework for understanding and predicting NRTI resistance. Such an approach could complement existing resistance testing, support more informed treatment decisions, and contribute to maintaining durable viral suppression and progress towards ending HIV as a public health threat.

## METHODOLOGY

### Dataset Structure & Feature Engineering

The dataset used consists of HIV-1 reverse transcriptase (RT) sequence-derived mutation profiles linked to drug-specific susceptibility measurements obtained from the Stanford HIV Drug Resistance Database (HIVDB).

Each record represents an individual HIV-1 isolate, with the viral RT mutation pattern serving as the predictor information and phenotypic resistance measurements for individual NRTIs serving as outcome variables.

The dataset is structured to enable supervised machine learning, where the objective is to learn relationships between RT amino acid mutation profiles and NRTI resistance phenotypes. Each row represents a unique HIV-1 isolate

Each column represents either:

- isolate metadata;
- RT mutation information;
- drug susceptibility measurements.

Data Cleaning Actions:

- Remove duplicate isolates
- Confirm HIV-1 subtype
- Verify RT sequence quality
- Remove incomplete records
- Standardize mutation notation.

IsolateID: unique identifier assigned to each HIV-1 sequence record.

Mutation\_List: contains the amino acid substitutions identified within the HIV-1 reverse transcriptase gene.

A binary mutation matrix generated.

*Original dataset:*

***IsolateID    Mutation\_List***

*A001        M184V,K65R*

*A002        M184V*

*A003        T215Y*

*Converted matrix:*

***IsolateID    M184V    K65R    T215Y***

*A001        1           1           0*

*A002        1           0           0*

*A003        0           0           1*

Where:

- 1 = mutation present
- 0 = mutation absent

Creates the feature matrix:

$X = \{x_1, x_2, x_3, \dots, x_n\}$

Each  $x_i$  represents an individual RT mutation.

Fold change values converted into binary resistance labels.

E.g:

| <b>Fold change</b> | <b>Label</b> |
|--------------------|--------------|
|--------------------|--------------|

|              |                    |
|--------------|--------------------|
| Below cutoff | 0 =<br>susceptible |
|--------------|--------------------|

|              |               |
|--------------|---------------|
| Above cutoff | 1 = resistant |
|--------------|---------------|

Allows supervised classification:

$X$  = RT mutations

$Y$  = NRTI resistance

## Supervised Learning

Supervised machine learning algorithms used to include Logistic Regression (LR), Random Forest (RF), Support Vector Matrix (SVM), and XGBoost to learn associations between HIV-1 RT mutation profiles and NRTI resistance outcomes.

Models evaluated using multiple performance metrics - accuracy, precision, recall, F1-score, Matthews Correlation Coefficient (MCC), ROC-AUC, and confusion matrices, to determine predictive performance and generalizability, and SHAP values to determine mutations which most influence predictions.

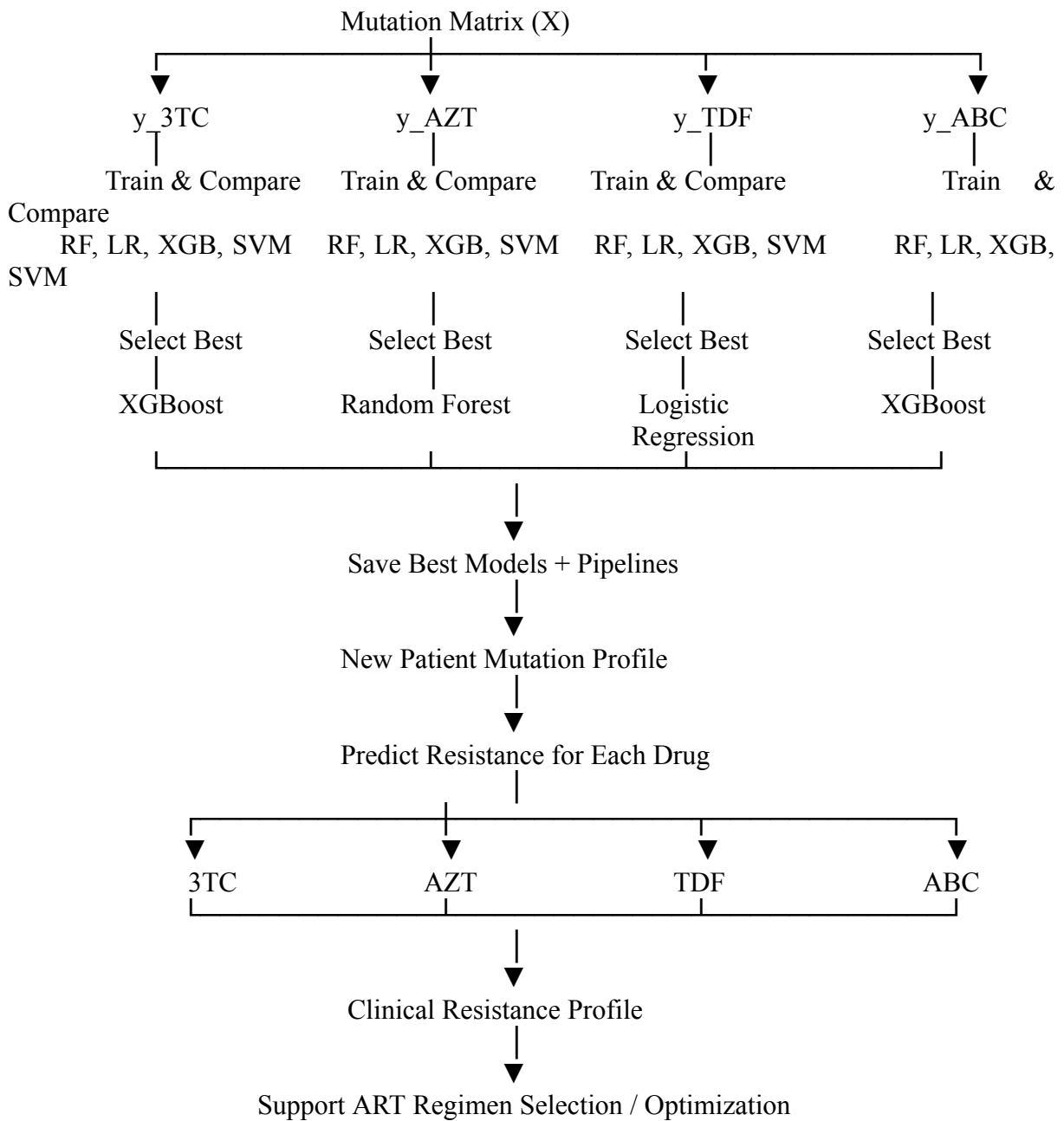
## Workflow

Describing ML architecture for predicting resistance to HIV-1 NRTIs using one mutation feature matrix and separate optimized models for each drug.

1. Load Stanford HIVDB dataset
2. Clean mutation strings
3. Build Mutation\_List
4. Use MultiLabelBinarizer to create a mutation feature matrix (X)
5. Create variable, y from resistance labels for each drug (3TC, AZT, TDF)
6. Explore with PCA and UMAP
7. Cluster with K-Means
8. Train multiple algorithms for each drug (Logistic Regression, Random Forest, SVM, and XGBoost)
9. Evaluate models using cross-validation and performance metrics
10. SHAP evaluation
11. Select the best model for each drug independently

12. Clinical application using one patient's mutation profile to generate predictions from drug-specific models.

## Architecture





## RESULTS

### IDENTIFICATION OF RT MUTATIONS ASSOCIATED WITH NRTI RESISTANCE USING THE SHDRDB

#### Data Selection

Phenotypic drug susceptibility data were obtained from the Stanford HIV Drug Resistance Database (SHDRDB) using the Detailed Phenotype Query interface. The search was restricted to HIV-1 isolates with reverse transcriptase (RT) gene sequences, with no codon restrictions applied to maximize the inclusion of available mutation profiles. Only susceptibility results generated using the PhenoSense phenotypic assay were included. Data were retrieved for all nucleoside reverse transcriptase inhibitors (NRTIs). Analysis focused only on lamivudine (3TC), tenofovir (TDF), abacavir (ABC), and zidovudine (AZT), drugs corresponding to those used in the first-line antiretroviral therapy (ART) regimen in Kenya. These data formed the basis for subsequent preprocessing, mutation profiling, and machine learning (ML) analyses of NRTI resistance.

#### Data Exploration and Cleaning

The initial dataset comprised 2,505 isolates and 335 variables. Following data exploration and preprocessing, the dataset was reduced to 2,505 isolates and 23 variables by removing non-informative and metadata fields that were not required for the predictive analyses. Excluded variables included: RefID, Species, Type, Method, NNRTIDRMs, CompleteMutationListAvailable, Author, NonDRMs, RefYear, MedlineID, Title, and PtID, as well as all protease inhibitor (PI)-related variables. Data quality assessment identified no duplicate Isolate IDs, confirming that each record represented a unique viral isolate. Empty mutation list fields were standardized by replacing missing entries with empty strings ("") to ensure consistent data formatting. Missing data were excluded to minimize their impact on subsequent analyses. *Figure 1* shows a summary of the drugs with missing data.

To further explore the underlying structure of the mutation data, an unsupervised machine learning approach was employed to identify natural groupings of HIV-1 reverse transcriptase isolates without using prior drug resistance labels. A total of 165 reverse transcriptase mutation features extracted from the SHDRDB were standardized before clustering. The optimal number of clusters was determined using the Elbow and Silhouette methods (*Figure 2*). The Elbow analysis demonstrated a clear inflection point at  $k = 2$ , indicating that additional clusters provided only marginal improvements in explaining the variation within the data. This finding was supported by the Silhouette score of 0.51, the highest among all evaluated values of  $k$ , indicating moderate-to-strong within-cluster cohesion and between-cluster separation. Based on these results, K-means clustering ( $k = 2$ ) was performed, successfully partitioning the HIV-1 isolates into two distinct, non-overlapping groups according to their mutation profiles.

To visualize the clustering structure, Principal Component Analysis (PCA) (*Figure 3a.*) was applied, revealing clear separation between the two clusters, thereby confirming their structural distinctness in reduced-dimensional space. This separation was independently validated using Uniform Manifold Approximation and Projection (UMAP) (*Figure 3b.*),

which reproduced the same two-cluster topology, suggesting that the observed clustering reflected genuine underlying biological structure rather than an artifact of dimensionality reduction. The identified clusters were subsequently interpreted based on their mutation profiles and compared with phenotypic NRTI drug resistance outcomes to determine whether naturally occurring mutation patterns were associated with differences in resistance. Collectively, this exploratory analysis demonstrated meaningful heterogeneity within the dataset and provided a robust foundation for the subsequent development and evaluation of supervised ML models for HIV-1 drug resistance prediction.

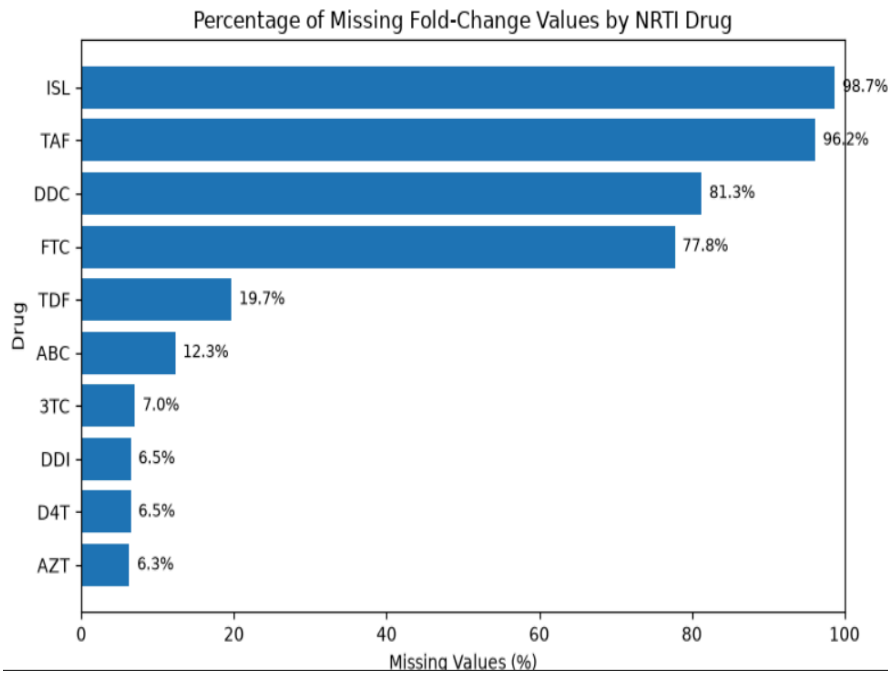


Figure 1: Showing percentage of missing values per drug

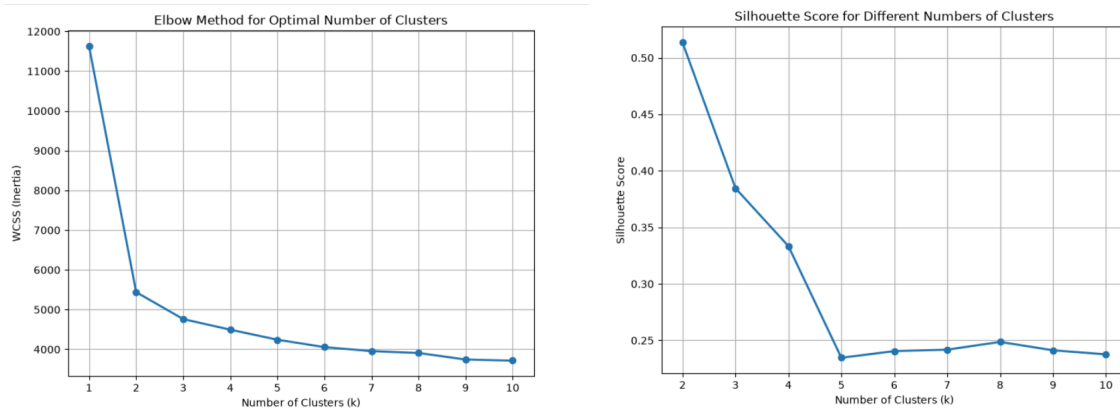
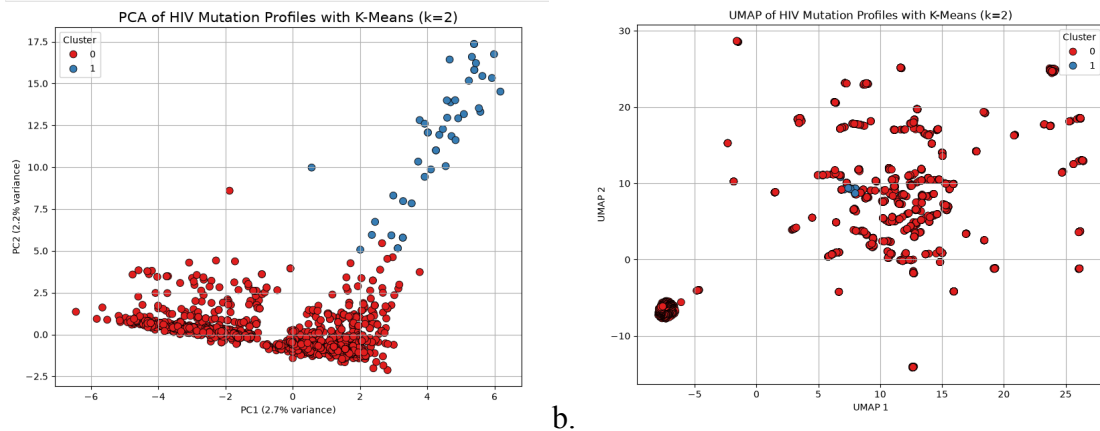


Figure 2: Elbow and Silhouette methods to optimise number of clusters



a. b.  
Figure 3: Mutation pattern visualisation by Uniform Manifold Approximation and Projection (UMAP).

Characterization of the mutation profiles within the identified clusters revealed two distinct NRTI resistance-associated patterns. Mutation Profile Cluster (MPC) 0 demonstrated a mutation signature with the most dominant discriminatory mutation being M184V (46.1%) and mutations consistent with a thymidine analogue mutation (TAM)-associated resistance profile to include M41L (40.4%), and T215Y (35.8%). These mutations are commonly associated with reduced susceptibility to multiple NRTIs such as 3TC and ABC (M184V). In contrast, MPC 2 exhibited a distinct Q151M/F116Y/K65R-S68G-associated resistance profile, characterized by high-frequency mutations including Q151M (100%), F116Y (95.6%), and V75I (73.9%), with additional enrichment of F77L (63.04%) and S68G (52.17%). The clusters highlighted in the heatmap (Figure 4). These cluster-specific mutation signatures indicate the presence of biologically distinct resistance pathways within the HIV-1 reverse transcriptase sequences and provide further evidence that the unsupervised clustering captured meaningful patterns of NRTI resistance-associated variation.

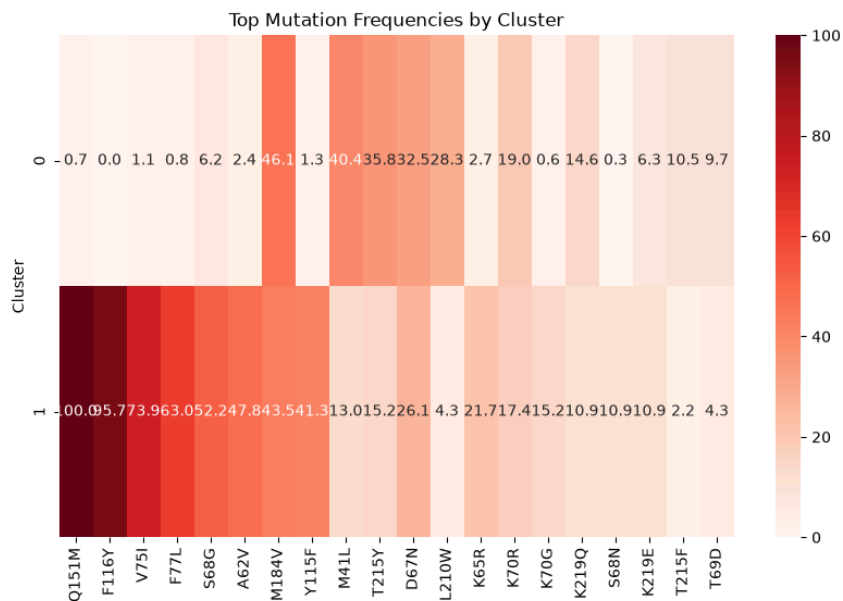


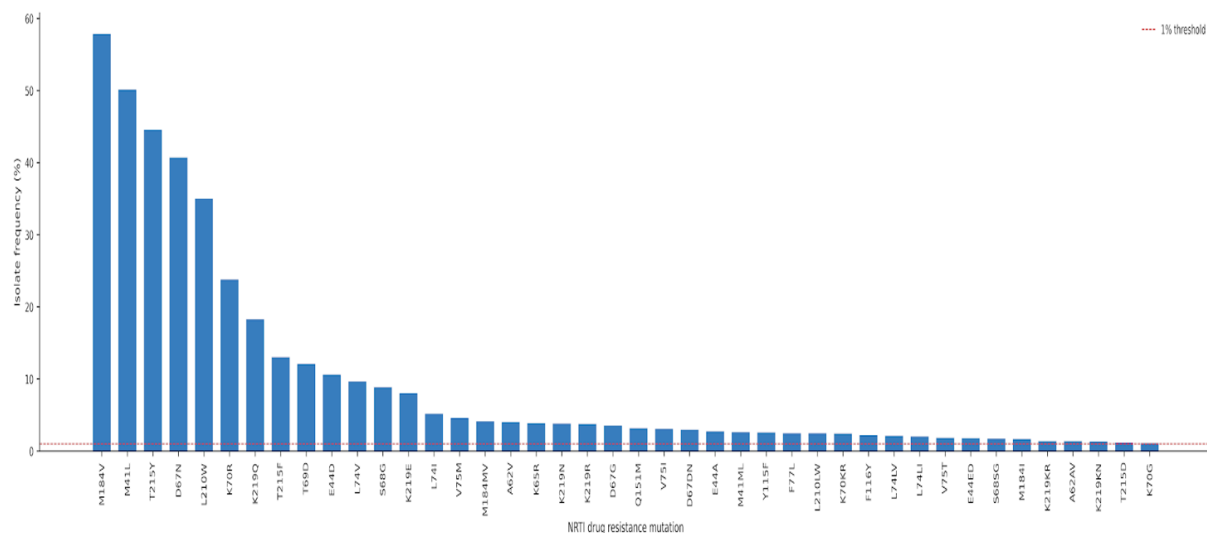
Figure 4: Heatmap showing mutation profile clusters observed

To assess the clinical relevance of the identified mutation clusters, Chi-square tests were performed to determine whether cluster membership was associated with phenotypic drug resistance outcomes. Significant associations were observed for all four evaluated NRTIs (3TC, AZT, TDF, and ABC;  $p < 0.001$  for each drug), indicating that the mutation-based clusters corresponded to distinct resistance phenotypes. Lamivudine (3TC) resistance was strongly associated with cluster assignment, likely reflecting the high prevalence of the M184V mutation in Cluster 0. Similarly, zidovudine (AZT) resistance showed a strong association with Cluster 0, consistent with the enrichment of thymidine analogue mutations (M41L, T215Y, and D67N). Tenofovir Disoproxil Fumarate (TDF) resistance was also significantly differentiated between clusters, suggesting distinct mutation profiles influencing susceptibility. In contrast, abacavir (ABC) resistance was significantly associated with Cluster 1, which demonstrated enrichment of the K65R-associated resistance pathway. These findings demonstrate that the unsupervised clustering approach identified mutation patterns that were significantly linked to clinically relevant NRTI resistance phenotypes.

## MODEL TRAINING

### Mutation Feature Selection by Frequency

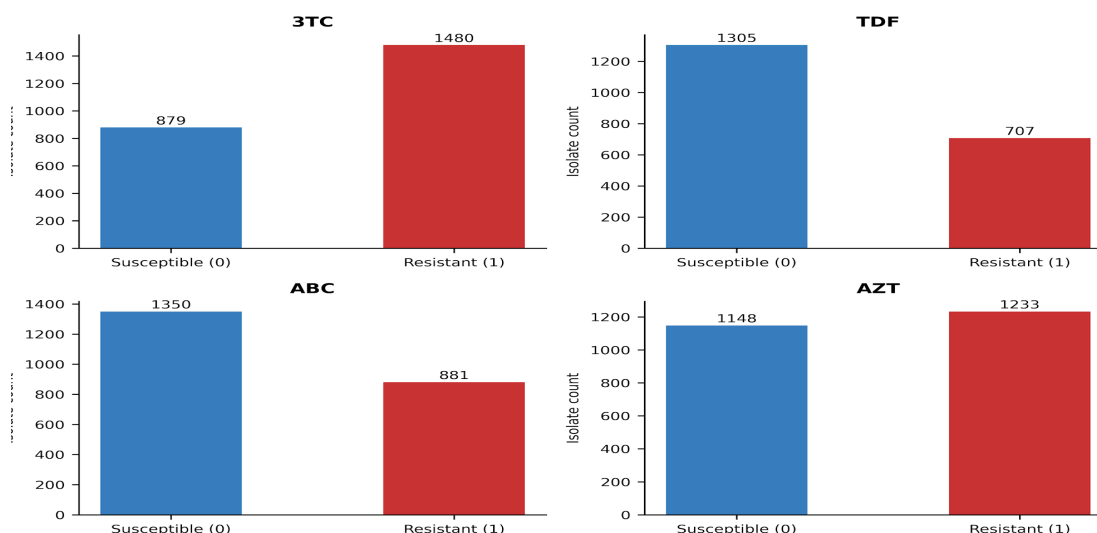
The first step in model development involved feature selection based on mutation frequency to reduce dimensionality and remove rare mutations that contribute little to predictive performance. The frequency distribution of reverse transcriptase mutations across the 2,505 HIV-1 isolates was calculated, and mutations present in at least 1% of isolates were retained for subsequent analyses. As illustrated in *Figure 4*, M184V was the most prevalent mutation, occurring in approximately 58% of isolates, followed by M41L (50%), T215Y (44%), K70R (41%), and L210W (35%). Other clinically important NRTI resistance mutations, including K65R, K219Q, T69D, E44D, L74V, and K219E, were also retained after exceeding the predefined frequency threshold. Mutations occurring in fewer than 1% of isolates were excluded from further analyses to reduce data sparsity, improve computational efficiency, and minimize noise while preserving mutations with established relevance to NRTI resistance. The resulting mutation set constituted the input features for clustering and subsequent supervised machine learning models.



*Figure 4. Frequency distribution of NRTI-associated reverse transcriptase mutations among HIV-1 isolates. The red dashed line indicates the 1% frequency threshold used for feature selection. Mutations above this threshold were retained for downstream clustering and predictive modeling.*

### Resistance Distribution (Class Imbalance)

Prior to model development, the distribution of phenotypic resistance classes was assessed for each of the four NRTIs to identify potential class imbalance. The distribution varied across the selected drugs (*Figure 5*). For 3TC, 1,480 isolates (62.8%) were classified as resistant compared with 879 (37.2%) susceptible isolates. TDF showed the opposite pattern, with 1,305 (64.9%) susceptible and 707 (35.1%) resistant isolates. ABC, 1,350 (60.5%) isolates were susceptible and 881 (39.5%) resistant, while AZT had a relatively balanced distribution, with 1,233 (51.8%) resistant and 1,148 (48.2%) susceptible isolates. These findings demonstrated varying degrees of class imbalance across the four drug-specific datasets and highlighted the need to account for class distribution during model development to minimize bias towards the more prevalent resistance class and ensure reliable prediction of both resistant and susceptible phenotypes.



*Figure 5: Resistance phenotype distribution and class imbalance across the four NRTIs. Distribution of HIV-1 isolates classified as susceptible (0) or resistant (1) to lamivudine (3TC), tenofovir (TDF), abacavir (ABC), and zidovudine (AZT). Bars represent the number of isolates in each resistance class, illustrating the differing degrees of class imbalance across the four drug-specific datasets prior to machine learning model development.*

## PREDICTIVE PERFORMANCE AND INTERPRETABILITY

### ROC-AUC Evaluation

Receiver operating characteristic (ROC) curves (*Figure 6*) were used to evaluate the ability of the four machine learning models tested – Logistic Regression (LR), Random Forest (RF), Support Vector Machine (SVM), and XGBoost (XGB) – to discriminate between resistant and susceptible isolates for each NRTI. All models demonstrated strong discriminatory performance with ROC-AUC values substantially above the 0.50 random-classification threshold. For 3TC, LR achieved the highest performance (AUC = 0.980), followed by SVM (0.977), XGB (0.972), and RF (0.962). According to the specific NRTIs, RF and XGB showed the highest AUC (0.967 each) for AZT, closely followed by LR (0.966) and SVM (0.959). For ABC, LR produced the highest AUC (0.944), with SVM (0.943), RF (0.940), and XGB (0.933) showing comparable performance. TDF yielded comparatively lower discrimination, although all models remained strong, with LR achieving the highest AUC (0.921), followed by SVM (0.909), XGB (0.900), and RF (0.878). Overall, the ROC-AUC analysis demonstrated that all four algorithms had good-to-excellent discriminatory ability across the four NRTIs, with 3TC showing the strongest overall predictive performance and TDF presenting the greatest challenge for classification.

## ROC-AUC

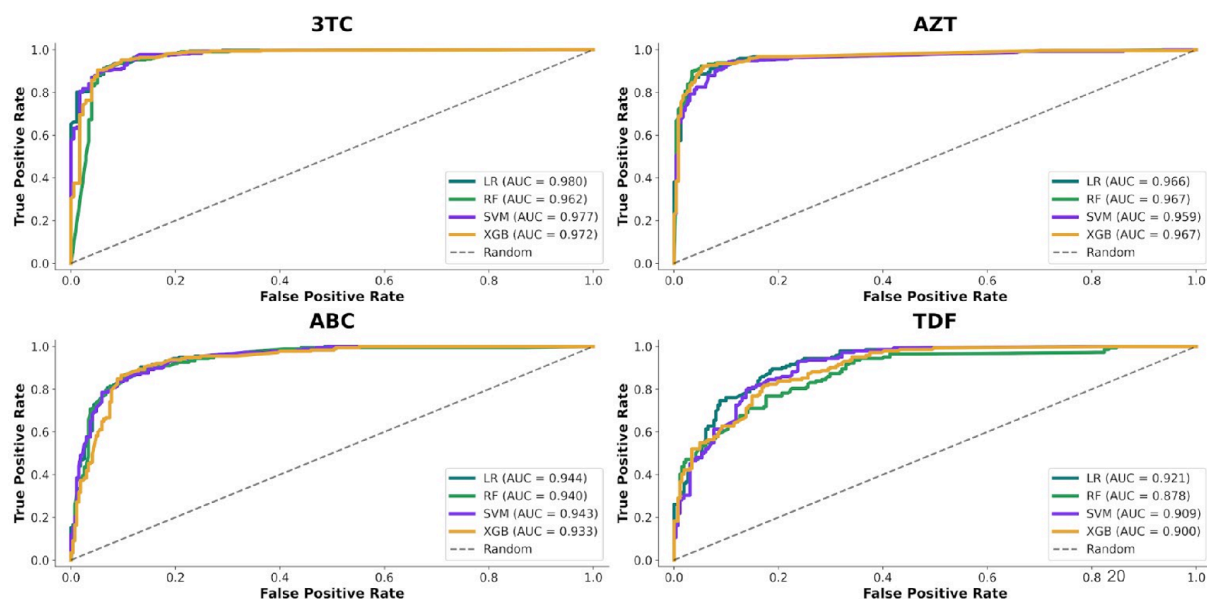


Figure 6: Receiver operating characteristic (ROC) curves comparing the discriminatory performance of Logistic Regression (LR), Random Forest (RF), Support Vector Machine (SVM), and XGBoost (XGB) models for predicting resistance to lamivudine (3TC), zidovudine (AZT), abacavir (ABC), and tenofovir (TDF). Area under the ROC curve (AUC) values are shown for each model.

## Precision-Recall Evaluation

Precision-recall (PR) curves were used as a complementary measure of model performance, particularly to assess the ability of the algorithms to correctly identify resistant isolates while minimizing false-positive classifications. Average precision (AP) was used as the summary measure of the area under each precision-recall curve. For lamivudine (3TC), all models demonstrated excellent performance, with LR achieving the highest AP (0.988), followed by SVM (0.986), XGB (0.980), and RF (0.962). For zidovudine (AZT), performance was similarly strong, with LR achieving an AP of 0.969, while RF and XGB both achieved 0.968, and SVM achieved 0.963. For abacavir (ABC), LR again showed the highest performance (AP = 0.907), followed closely by SVM (0.905), RF (0.893), and XGB (0.876). Tenofovir (TDF) demonstrated comparatively lower precision-recall performance, with LR achieving the highest AP (0.857), followed by XGB (0.832), SVM (0.826), and RF (0.816). Overall, the PR analysis was consistent with the ROC-AUC findings, demonstrating strong model performance for 3TC and AZT, moderate-to-strong performance for ABC, and comparatively reduced performance for TDF. Across the four drugs, LR consistently achieved the highest AP, indicating strong precision and sensitivity in identifying resistant isolates.

## Precision-Recall

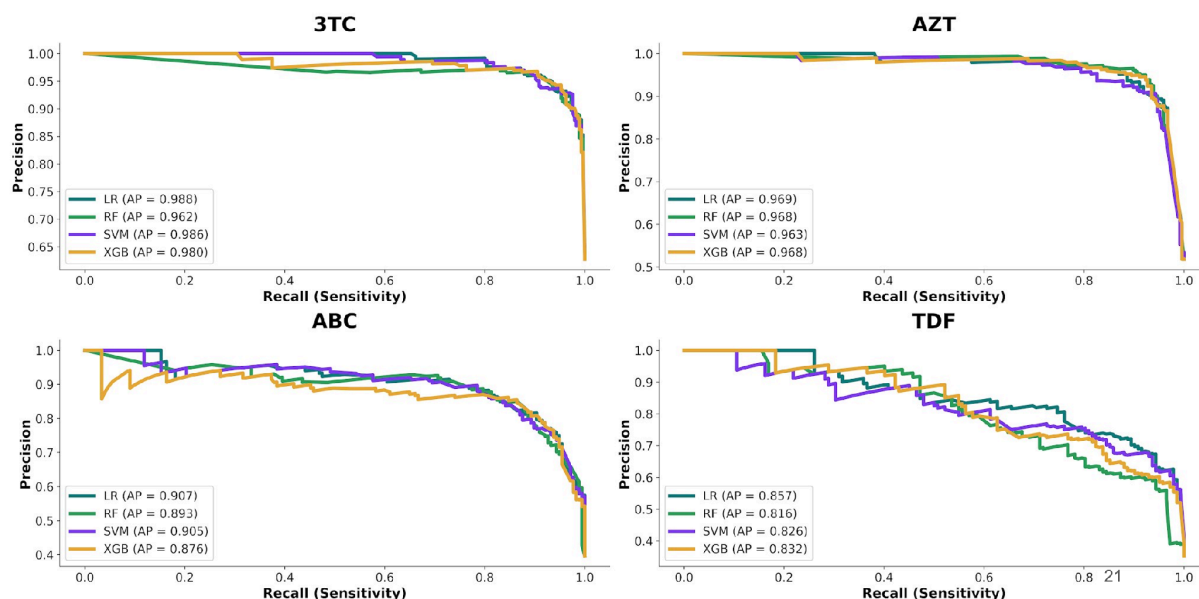


Figure 7: Precision-recall curves comparing the performance of Logistic Regression (LR), Random Forest (RF), Support Vector Machine (SVM), and XGBoost (XGB) models in predicting NRTI resistance. Average precision (AP) values are shown for each model and drug, providing an assessment of precision and sensitivity across classification thresholds.

## F1 Scores

The F1-score, a performance metric, was calculated to harmonise precision and recall and evaluate our models (Table 1). Across the four drugs, model performance varied according to the drug being predicted. For 3TC, XGBoost (XGB) demonstrated the strongest performance, achieving the highest F1-score of 0.93, with a precision of 0.93 and recall of 0.92. For ABC, XGB also performed best, achieving a precision, recall, and F1-score of 0.88, compared with F1-scores of 0.87 for LR, RF, and SVM. In contrast, TDF showed lower overall predictive performance, with LR achieving the highest F1-score (0.83) and recall (0.85), while RF, SVM, and XGB achieved F1-scores of 0.78, 0.81, and 0.80, respectively. For AZT, RF and XGB produced the highest F1-score (0.93); however, RF achieved a slightly higher recall (0.94) than XGB (0.93), while both had a precision of 0.93. Overall, XGB showed the most consistently strong performance across the four drugs, achieving the highest or joint-highest F1-score for 3TC, ABC, and AZT, whereas LR performed best for TDF.

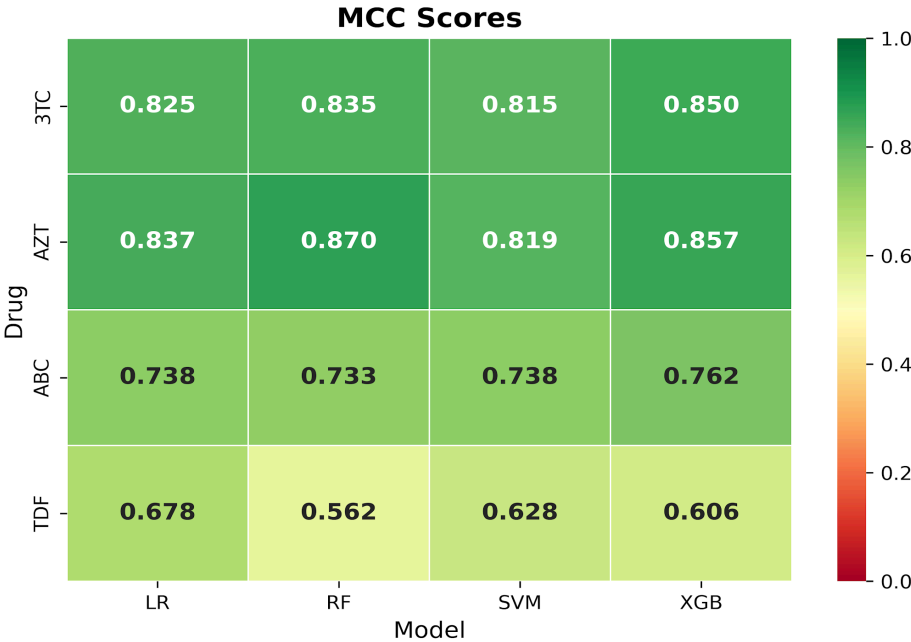


| Drug | Model | Precision | Recall | F1_score |
|------|-------|-----------|--------|----------|
| 3TC  | LR    | 0.91      | 0.92   | 0.91     |
| 3TC  | RF    | 0.91      | 0.92   | 0.92     |
| 3TC  | SVM   | 0.9       | 0.91   | 0.91     |
| 3TC  | XGB   | 0.93      | 0.92   | 0.93     |
| ABC  | LR    | 0.87      | 0.87   | 0.87     |
| ABC  | RF    | 0.86      | 0.87   | 0.87     |
| ABC  | SVM   | 0.87      | 0.87   | 0.87     |
| ABC  | XGB   | 0.88      | 0.88   | 0.88     |
| TDF  | LR    | 0.83      | 0.85   | 0.83     |
| TDF  | RF    | 0.78      | 0.79   | 0.78     |
| TDF  | SVM   | 0.8       | 0.83   | 0.81     |
| TDF  | XGB   | 0.8       | 0.81   | 0.8      |
| AZT  | LR    | 0.92      | 0.92   | 0.92     |
| AZT  | RF    | 0.93      | 0.94   | 0.93     |
| AZT  | SVM   | 0.91      | 0.91   | 0.91     |
| AZT  | XGB   | 0.93      | 0.93   | 0.93     |

*Table 1: Performance of machine learning models for drug prediction across 3TC, ABC, TDF, and AZT, showing precision, recall, and F1-score for logistic regression (LR), random forest (RF), support vector machine (SVM), and XGBoost (XGB).*

The Matthews correlation coefficient (MCC) was used as an additional measure of classification performance. MCC is a balanced metric, particularly useful when the classes are imbalanced, that evaluates the quality of binary classifications, taking into account true positives, true negatives, false positives, and false negatives. For 3TC, all models achieved high MCC values, with XGB showing the highest score (0.850), followed by RF (0.835), LR (0.825), and SVM (0.815). For AZT, RF achieved the highest MCC (0.870), followed by XGB (0.857), LR (0.837), and SVM (0.819). For ABC, MCC values were comparatively lower, ranging from 0.733 for RF to 0.762 for XGB, while both LR and SVM achieved 0.738. TDF demonstrated the lowest MCC values across all four drugs, with LR performing best (0.678), followed by SVM (0.628), XGB (0.606), and RF (0.562). Overall, the MCC results indicate strong classification performance for 3TC and AZT, moderate performance for ABC, and comparatively reduced performance for TDF. Unlike the PR analysis, where LR

consistently achieved the highest AP, MCC performance varied across models, with XGB performing best for 3TC and ABC, RF for AZT, and LR for TDF. This variation suggests that model performance differed according to the drug, with no single algorithm consistently outperforming the others across all datasets.

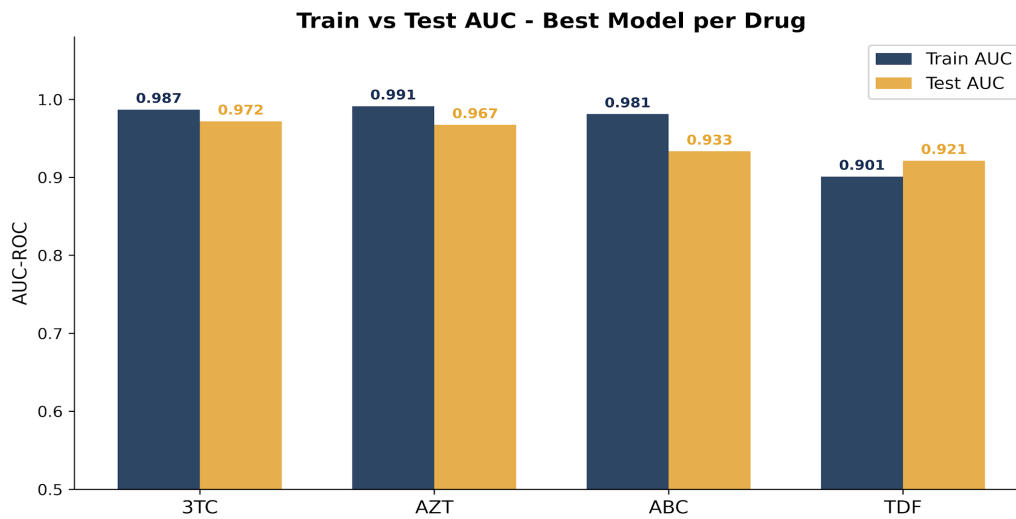


*Figure 8. Matthews correlation coefficient (MCC) scores for the evaluated machine learning models across the four antiretroviral drugs. MCC values are shown for Logistic Regression (LR), Random Forest (RF), Support Vector Machine (SVM), and Extreme Gradient Boosting (XGB) models. Higher MCC values indicate better overall classification performance, with the results demonstrating generally strong performance for 3TC and AZT, moderate performance for ABC, and comparatively lower performance for TDF.*

**Selection of Best Model for each NRTI**

To assess model generalizability and potential overfitting, the AUC-ROC values obtained on the training and test datasets were compared for the best-performing model for each drug. For 3TC, the training AUC was 0.987 compared with a test AUC of 0.972, representing a small decrease of 0.015. Similarly, for AZT, the training AUC of 0.991 decreased to 0.967 on the test dataset, corresponding to a difference of 0.024. For ABC, the training AUC was 0.981, while the test AUC was 0.933, resulting in a larger decrease of 0.048. In contrast, TDF showed a test AUC of 0.921 compared with a training AUC of 0.901, with the test performance exceeding the training performance by 0.020. Overall, the relatively small differences between training and test AUCs for 3TC and AZT suggest good model generalizability with limited evidence of overfitting, whereas the larger reduction observed for ABC may indicate some degree of overfitting or greater variability in model performance. The comparable test performance for TDF, despite its lower training AUC, suggests that the model retained good discriminatory ability on unseen data. Overall, these findings indicate that the best-performing models generally maintained strong performance when evaluated on

the test datasets.



*Figure 9. Train versus test AUC-ROC for the best-performing model for each antiretroviral drug. Bars show AUC-ROC values obtained on the training and test datasets for 3TC, AZT, ABC, and TDF, illustrating the degree to which model discrimination was maintained on unseen data.*

## SHAP Analysis

SHapley Additive Explanations (SHAP) analysis was conducted to exam how specific reverse RT influence the model's predicted resistance phenotype. The SHAP summary plots (*Figure 10*) illustrate the contribution of individual HIV-1 RT mutations to the predictions of the ML models for NRTI resistance. Across the 3TC, AZT, ABC, and TDF models, mutations such as L210LW, M184I, K70KR, Q151M, K219R, and S68SG consistently appear among the most influential features, although their magnitude and direction of contribution vary between drugs and models. In the 3TC model, L210LW shows particularly strong positive SHAP values, indicating that this mutation is strongly associated with increased predicted resistance, while M184I and K70KR also contribute substantially. The AZT model similarly identifies M184I, K70KR, K219R, and S68SG as important contributors, with several mutations producing positive SHAP values associated with increased resistance predictions. For ABC, L210LW and M184I are among the dominant features, with L210LW demonstrating a clear separation between negative and positive contributions. The TDF model shows a somewhat different pattern, with L210LW, K219R, A62V, and M41L producing notable effects, including both positive and negative contributions.

Overall, the plots demonstrate that the models rely on biologically relevant mutation patterns rather than treating all mutations equally, and that the influence of individual mutations is drug-specific and dependent on the predicted resistance phenotype. The spread of SHAP values also indicates heterogeneity in mutation effects across samples, highlighting potential interactions between mutations and the importance of considering combinations of

resistance-associated substitutions when predicting HIV-1 NRTI resistance.

## SHAP Summary Plot

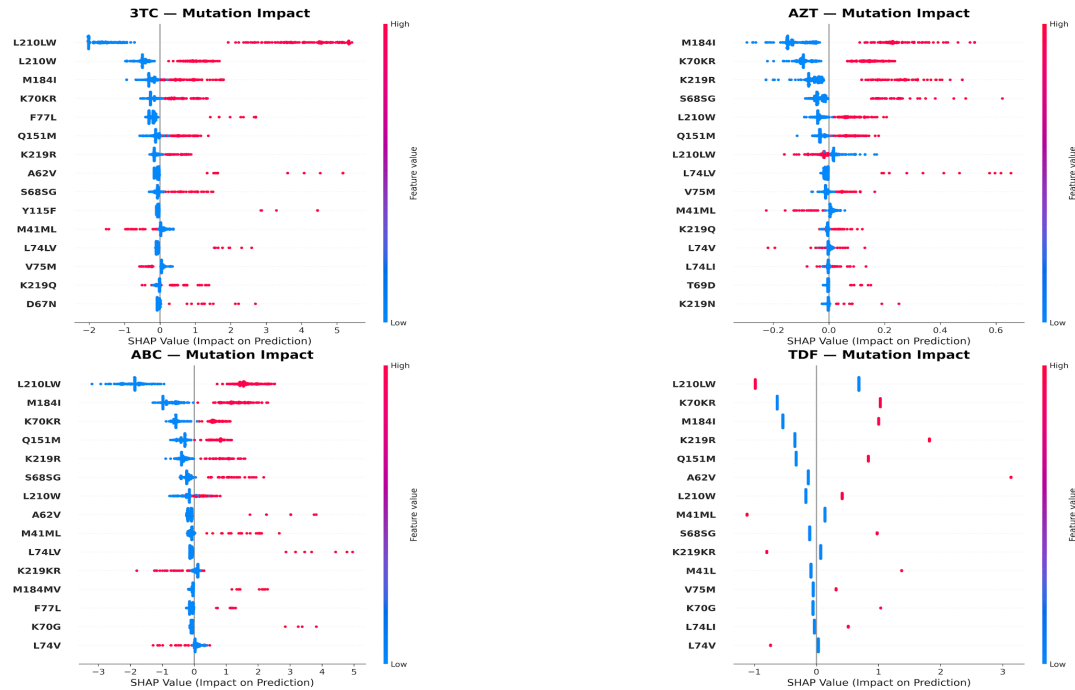


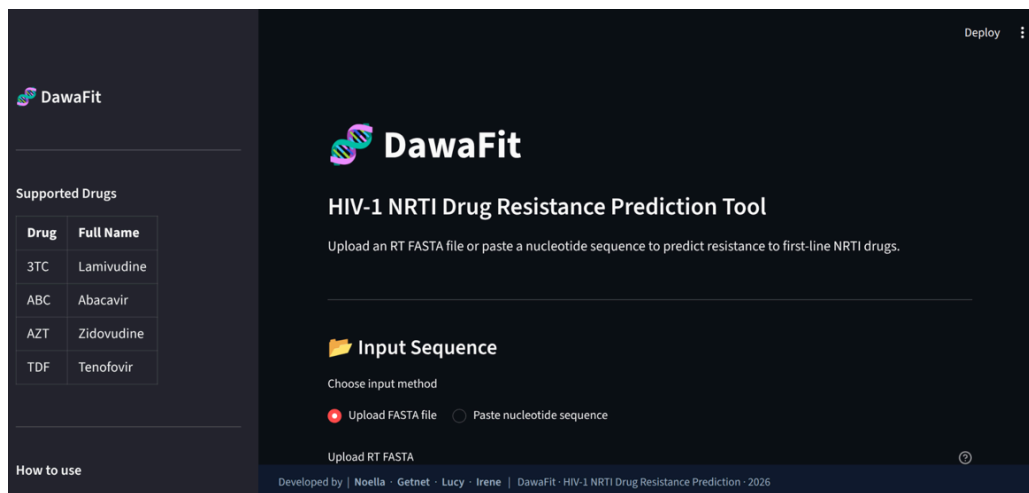
Figure 10: SHAP summary plots illustrating the contribution of HIV-1 reverse transcriptase mutations to machine learning predictions of NRTI resistance. Each panel represents a model trained to predict resistance to 3TC, AZT, ABC, or TDF. Mutations are ranked according to their overall importance, with each point representing an individual sample and its SHAP value indicating the magnitude and direction of the mutation's contribution to the model prediction. Positive SHAP values indicate a contribution toward increased predicted resistance, whereas negative values indicate a contribution toward decreased predicted resistance. Point colour represents the mutation feature value, with red indicating higher feature values and blue indicating lower values. The plots highlight drug-specific differences in the relative importance and effects of resistance-associated mutations.

## INTEGRATION & DEPLOYMENT

### DawaFit Application

The development of machine learning models for HIV-1 NRTI drug resistance prediction has potential clinical implications for supporting more informed and individualized antiretroviral therapy (ART) selection. By predicting resistance to first-line NRTIs, these models may help identify drugs that are less likely to be effective against a patient's viral strain, thereby supporting the selection of more appropriate treatment options and potentially reducing the risk of treatment failure. To translate the modelling results into a practical application, the ResiMap DawaFit HIV-1 NRTI Drug Resistance Prediction Tool was developed [[Resimap DawaFit](#)].

DawaFit provides a user-friendly interface through which users can upload an RT FASTA file or paste a nucleotide sequence and obtain predictions of resistance to the supported NRTIs, including lamivudine (3TC), abacavir (ABC), zidovudine (AZT), and tenofovir (TDF). This application demonstrates the potential for integrating machine learning-based resistance prediction into a practical decision-support platform, particularly in settings where rapid interpretation of HIV-1 sequence data could complement conventional drug-resistance testing. However, DawaFit should be considered a decision-support tool rather than a replacement for established genotypic resistance testing and clinical judgement, and further external validation using diverse patient populations and clinical datasets would be required before routine clinical implementation.



*Figure 11. ResiMap DawaFit HIV-1 NRTI Drug Resistance Prediction Tool landing page. The interface provides users with options to upload an RT FASTA file or paste a nucleotide sequence for prediction of resistance to selected first-line NRTI drugs, including 3TC, ABC, AZT, and TDF.*

## DISCUSSION

The mutation patterns identified are broadly consistent with NRTI resistance-associated mutations previously documented among HIV-1-infected populations in Kenya, supporting the biological relevance of the mutation clusters identified in our analysis. Particularly, M184V/I, K65R, and the thymidine analogue mutations (TAMs) M41L, D67N, K70R, L210W, T215Y/F, and K219Q/E have been reported in Kenyan cohorts, including treatment-experienced individuals and patients experiencing virological failure. M184V/I has consistently been reported as the predominant NRTI resistance mutation in Kenya and is strongly selected by 3TC exposure, whereas K65R is primarily associated with tenofovir-containing regimens. *Table 2* summarises most consistently reported NRTI resistance mutations in Kenya.

The resistance mechanisms represented by these mutations broadly involve two processes: reduced incorporation (nucleotide discrimination), whereby mutant RT preferentially incorporates natural nucleotides over NRTI analogues, and ATP-mediated excision (primer unblocking), in which the incorporated analogue is removed from the nascent viral DNA strand, allowing reverse transcription to resume. The latter mechanism is characteristic of TAMs, including the TAM-1 pathway (M41L, L210W, and T215Y) and TAM-2 pathway (D67N, K70R, and K219Q/E). The predominance of M184V in Cluster 0 and the presence of TAM-associated mutations therefore mirror resistance patterns described in Kenya, while the K65R-associated profile observed in Cluster 1 is also consistent with documented Kenyan NRTI resistance. These concordant patterns suggest that the clustering approach captured biologically meaningful resistance pathways rather than arbitrary statistical groupings, strengthening the interpretation of the identified clusters and their associations with the observed NRTI resistance phenotypes.

| Mutation       | Commonly Associated Drugs                                                                                                                                                                                                                                                                                                           | Mechanism                                                                                                                                | Notes                                                                                                                                                                                             | Reference                                                                                                                                                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>M184V/I</b> | <p>Lamivudine (3TC), Emtricitabine (FTC)</p> <p>-Produces low-level resistance to ABC and DDI (didanosine).<br/> - Reduces viral replication capacity (fitness cost).<br/> - Increases susceptibility to AZT and TDF, explaining why clinicians may continue 3TC despite resistance because the mutation weakens viral fitness.</p> | <p>Discrimination mutation.</p> <p>Alters the nucleotide-binding pocket of RT, reducing incorporation of 3TC and FTC into viral DNA.</p> | <p>-Most prevalent NRTI mutation in treatment-experienced patients.</p> <p>-Nairobi study: NRTI primary mutations M184V/I and K65R/E/N were found in 28.8% and 8.9% of subjects respectively.</p> | <p>Scriven, Y. A., <i>et al.</i> (2021). Prevalence and factors associated with HIV-1 drug resistance mutations in treatment-experienced patients in Nairobi, Kenya: A cross-sectional study. <i>Medicine</i>, 100(40), e27460.</p> |

|                |                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                           |                                                                                                                                                                 |                                                                                                                                                                                                                                                                |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                |                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                           |                                                                                                                                                                 |                                                                                                                                                                                                                                                                |
| <b>K65R</b>    | <p>Tenofovir (TDF),<br/>Abacavir (ABC)</p> <p>-Intermediate resistance to 3TC, FTC<br/>- Usually co-occurs with M184V after failure of TDF/3TC-containing regimens.<br/>- Rarely occurs together with multiple TAMs because the two pathways are antagonistic.</p> | Discrimination mutation that decreases incorporation of multiple NRTIs                                                                                                                                                                                                    | <p>Increasingly reported following adoption of TDF-containing regimens.</p> <p>Reported in Nairobi cohort (8.9%) and among patients failing first-line ART.</p> | <p>Koigi, P., Ngayo, M.O., Khamadi, S. <i>et al.</i> HIV type 1 drug resistance patterns among patients failing first- and second-line antiretroviral therapy in Nairobi, Kenya. <i>BMC Res Notes</i> 7, 890 (2014).</p> <p>Scriven, Y. A., et al. (2021).</p> |
| <b>M41L</b>    | Zidovudine (AZT)                                                                                                                                                                                                                                                   | <p>Thymidine Analogue Mutation (TAM) 1 pathway</p> <p><i>TAMs promote ATP-dependent excision (primer unblocking). Instead of preventing drug incorporation, they remove the incorporated AZT molecule from viral DNA, allowing reverse transcription to continue.</i></p> | Documented as one of the major thymidine analogue mutations (TAMs) in Kenyan patients.                                                                          | Kinyua JG, <i>et al.</i> , Antiretroviral resistance among HIV-1 patients on first-line therapy attending a comprehensive care clinic in Kenyatta National Hospital, Kenya: a retrospective analysis. <i>Pan Afr Med J.</i> 2018 Apr 2;29:186.                 |
| <b>L210W</b>   | AZT                                                                                                                                                                                                                                                                | TAM 1 pathway                                                                                                                                                                                                                                                             | Less frequent but documented among patients in Nairobi.                                                                                                         | Scriven, Y. A., et al. 2021                                                                                                                                                                                                                                    |
| <b>T215Y/F</b> | AZT                                                                                                                                                                                                                                                                | TAM 1/2 pathway                                                                                                                                                                                                                                                           | Major TAM associated with prolonged treatment failure.                                                                                                          | Koigi, P., et al. 2014                                                                                                                                                                                                                                         |
| <b>D67N</b>    | AZT                                                                                                                                                                                                                                                                | TAM 2 pathway                                                                                                                                                                                                                                                             | Frequently reported TAM. Nairobi treatment-experienced patients.                                                                                                | Kinyua JG, et al., 2018                                                                                                                                                                                                                                        |
| <b>K70R</b>    | AZT                                                                                                                                                                                                                                                                | TAM 2 pathway                                                                                                                                                                                                                                                             | Often accompanies other TAMs.                                                                                                                                   | Koigi, P., et al. 2014                                                                                                                                                                                                                                         |
| <b>K219Q/E</b> | AZT                                                                                                                                                                                                                                                                | TAM 2 pathway                                                                                                                                                                                                                                                             | Accessory TAM                                                                                                                                                   | Scriven, Y. A., et al.                                                                                                                                                                                                                                         |

|  |  |  |                                                       |      |
|--|--|--|-------------------------------------------------------|------|
|  |  |  | reported in a Nairobi treatment-experienced patients. | 2021 |
|--|--|--|-------------------------------------------------------|------|

*Table 2: Summary of most consistently reported NRTI resistance mutations in Kenya.*

## CONCLUSION

This work demonstrates the potential of using machine learning (ML) to profile HIV-1 RT mutations and thus characterize and predict NRTI drug resistance. Unsupervised analyses identified distinct mutation clusters that were significantly associated with phenotypic resistance to 3TC, TDF, ABC, and AZT, with the observed mutation patterns, including M184V/I, K65R, and thymidine analogue mutations (TAMs), being broadly consistent with NRTI resistance profiles previously reported in Kenya. This suggests that the identified clusters represent biologically meaningful resistance pathways rather than arbitrary statistical groupings. Supervised ML models subsequently demonstrated strong discriminatory performance across the four drugs, as supported by both ROC-AUC and precision-recall analyses. Importantly, combining predictive modelling with identification of resistance-associated mutations provides both predictive capability and biological insight, while improving model transparency and interpretability. Collectively, this work supports the potential utility of interpretable ML as a complementary approach for HIV-1 drug-resistance surveillance and prediction, with potential to contribute to earlier identification of resistance and more informed treatment decisions. However, validation using independent and geographically representative datasets, particularly from Kenyan populations, remains necessary before translation into clinical or public health applications.



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